

Automatic compliance checking of BIM models against quality standards based on ontology technology

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ABSTRACT

Many countries have issued standards to stipulate the quality of BIM models. However, due to the complexity of BIM models and the textual expression of the standards, checking the quality of BIM models against the standards manually is challenging and time-consuming. In order to solve the problem, this paper presents an ontology-based approach for the automatic compliance checking of BIM models against the standards. Taking the Chinese standard as an example, it is represented in OWL to obtain an ontology, and SWRL is used for representing the checking rules. Then, an inference process is established to fulfill the checking. Finally, a prototype system is developed to validate the approach. It is concluded that the approach can be applied to check the quality of BIM models in IFC format automatically. This research contributes to the development of automatic compliance checking systems for the quality of BIM models against real standards.

1. Introduction

The rapid development of Building Information Modelling (BIM) technology has a huge impact on Architectural, Engineering and Construction (AEC) industry. In the application of BIM technology, a BIM model is not only the carrier for the information sharing among different BIM authoring software, but also the data basis in various BIM-based application scenarios [1]. Apparently, to ensure the effective application of BIM technology, BIM models should meet some requirements on data completeness, consistency and correctness [2]. For example, in the case of BIM-based automatic cost estimation, the input BIM model must contain complete and correct geometric and material attributes for all components, such as slabs and beams [3]. In this paper, the performance of BIM models on the three aspects of data, i.e., completeness, correctness and consistency, is collectively defined as the quality of BIM models.

Up to now, the quality of BIM models often cannot meet the requirements [2], which has impeded the fully exploitation of BIM technology greatly. In order to solve the problem, many countries and institutions have formulated relevant standards, such as Singapore BIM guide [4] and NBS BIM object standard [5]. In China, the “National Standards for Graphic Expression of Building Information Modelling (JGJ/T 448-2018)” (hereafter referred to as “the Chinese standard”) formulated by the Ministry of Housing and Urban-Rural Development

has been issued, which stipulates the requirements on the quality of BIM models for design delivery processes [6]. However, the BIM model of actual buildings usually includes a lot of components and is complex in structure; meanwhile, such standards consist of textual expressions and involve lots of requirements. Therefore, it is rather challenging and time-consuming to check the quality of BIM models in a manual way. Hence, it is necessary to automate the compliance checking of BIM models against quality standards.

Although there is a critical need for automating the compliance checking of BIM models against quality standards, no research has been reported to devote directly to the topic up to now. In order to solve the problem, obviously, two issues need to be addressed in advance: first, what format of BIM models should be targeted? Second, what approach should be used to deal with the quality standards?

As far as the first issue is concerned, it is well known that, over the past two decades, the Industry Foundation Classes (IFC) standard developed by the buildingSMART International have been widely used to store and exchange BIM models. Its latest version is IFC 4.3.0.0 and its computer interpretable version is presented by using EXPRESS language, in which entities are specified as the basic units to represent and organize building information. Considering that it has been accepted as an ISO (International Standardization Organization) standard [7] and

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supported by major vendors of BIM authoring software, it is feasible to focus on the quality of BIM models in IFC format (hereafter referred to as “IFC models”) to cope with the different formats of BIM models due to different BIM authoring software.

As far as the second issue is concerned, as will be explained in the “Literature Review” section, ontology technology is a promising technology to use, considering the potential of the technology and the recent trends for dealing with compliance checking problems.

Hence, this study aims to propose an automatic compliance checking approach for the quality of IFC models based on ontology technology. It deserves to explain that automatic compliance checking means to check a model against standards by using a computerized process [8]. In the approach, ontology technology is applied to represent the quality standard for BIM models in a computer-readable format and a rule engine is used to execute the checking automatically. There are two research questions for this study, i.e., how to represent the quality standard of BIM models by using ontology technology? How to implement a systematic approach to solve the problem? In response to the questions raised, the objectives of this study are as follows: (1) Realizing the computerized representation of Chinese standards based on ontology; (2) Establishing an automatic compliance checking method for BIM models; (3) Developing the corresponding prototype system according to the proposed method.

In this paper, the Chinese standard is used for presenting the process and implementation of the proposed approach. Considering the similarity and consistency of various quality standards for BIM models, the approach is also applicable for other similar quality standards as well. In the following, Section 2 carries out literature review; Section 3 presents the framework of the approach; Section 4 analyzes the contents of the Chinese standard for the quality of BIM models; Section 5 represents the Chinese standard by establishing an ontology in OWL (Web Ontology Language), called BIM-QC ontology (BIM model Quality Checking ontology), and the corresponding checking rules in SWRL (Semantic Web Rule Language) to form a ruleset; Section 6 deals with the acquisition of the instance of the BIM-QC ontology from an IFC model to carry out the automatic compliance checking; Section 7 illustrates the prototype system that we developed corresponding to the approach and its case study; and Section 8 gives the conclusions.

2. Literature review

Although many approaches have been proposed in the field of automatic compliance checking, most of them have focused on the design standards or construction standards and no study was found to be directly related to the automatic compliance checking for BIM models against quality standards. Thus, three highly relevant topics are reviewed in this section, i.e., quality checking for BIM models, automatic compliance checking and Linked Building Data (LBD), in order to identify the research gap.

2.1. Quality checking for BIM models

The existing researches related to quality checking for BIM models mainly focus on two topics, i.e., What to check regarding the quality of BIM models? How to check the quality of BIM models automatically?

2.1.1. Quality checking classification for BIM model

About the first topic, Solihin et al. [2] proposed four categories of automatic checking for the quality of BIM models, including grammatical correctness, semantic correctness, consistency, and completeness. On this basis, the rules for quality checking of IFC models were extracted and classified into four types, i.e., rules for data structure or syntax checking, that for semantic checking, that for Model View Definition (MVD) consistency checking, and that for import or export sanity checking [8]. Lee et al. [9] decomposed the quality checking of IFC models into three categories, i.e., syntactic compliance checking against

the IFC schema, semantic and syntactic conformity checking against MVD, and quality checking against application requirements.

2.1.2. Quality checking approaches for BIM model

About the second topic, several studies proposed rule-based approaches to check the quality of BIM models for specific aspects automatically. Lee et al. [10,11] established a computer-readable rule base for MVD conformity checking and developed a prototype system to execute the rules based on the IfcDoc tool. Fahad et al. [12] established a method to check the semantic error of IFC models, in which, IFC models were converted into the format of Resource Description Framework (RDF), a few new vocabularies were then created to represent the items to be checked, and a rule base was established and validated. Zhang et al. [13] developed a prototype system for MVD consistency checking with the support of an XML (eXtensible Markup Language)-format rule base computerized from the application requirements in an actual project. Jiang et al. [14] selected the BIM model of power grid engineering as the research object, developed a BIM model quality checking system based on knowledge graph, and created corresponding checking rules to verify whether it conforms to the relevant norms and standards of power grid engineering. According to the relevant standards, Zhao et al. [15] built an ontology suitable for the checking of the completeness of the subway station building information model, and realized the knowledge organization and representation of the subway building delivery area. At the same time, according to the created SWRL rules, the completeness checking of the design delivery stage model is realized by combining the IFC example file.

Furthermore, Ying et al. [16] established a rule base to check the completeness and consistency of the spatial boundary information in IFC models. Luttun et al. [17] presented an approach based on Boolean operations to check the geometric errors in IFC models, in which, the Boolean operation was used to calculate the relationship between geometric elements and it is compared with the correct geometric relationship. Abualdenien et al. [18] proposed a method to check the Level of Geometry (LOG) of components in BIM models, in which a set of features were defined to characterize the geometric accuracy of BIM components and the random forest algorithm was applied to evaluate the LOG of BIM components.

2.2. Automatic compliance checking

Eastman et al. proposed a general process for the automatic compliance checking based on BIM models as a rule-based one with four stages [19] for the first time, as shown in Fig. 1, where the Stage 1 is carried out only once for a specific standard.

Stage 1. Rule representation. In this stage, the contents of the standards are analyzed into rules and then computerized into a computer-readable language. Rule representation is the starting point and the core task of the automatic compliance checking process.

Stage 2. BIM model data preparation. In this stage, based on the rules in the standard, the required information is extracted from BIM models to form a subset of data. Then, the subset of data is converted into the format supporting inference.

Stage 3. Rule execution. In this stage, the rules represented from standards are executed on the BIM models for specific checking tasks.

Stage 4. Checking results reporting. In this stage, the results of compliance checking are shown to users.

Based on the process, semantic web technology was introduced into the automatic compliance checking in recent years, which provides a knowledge representation approach to enable computers to understand the semantics of standards and master domain knowledge [20]. In fact, ontology-based approaches have received great attention in various compliance checking tasks in design, construction, and maintenance phases. For the design phase, Ma et al. [21] applied ontology technology to the computerized representation of cost estimation standards, and further developed an automatic cost estimation system. Pauwels et al.

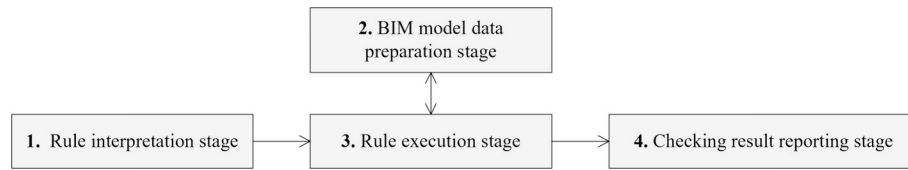


Fig. 1. General process of automatic compliance checking, proposed by Eastman et al [19]

[22,23] proposed a method to evaluate and inspect acoustic performance of buildings, in which the contents of relevant standards have been computerized into a computer-readable format by using RDF. For the construction phase, Zhong et al. [24] designed an ontology for the construction quality inspection of projects. Lu et al. [25] used OWL to computerize construction safety standards, and divided the concepts in construction safety standards into five categories: procedure, task, condition, hazard, and solution, and then formed a Construction Safety Checking Ontology (CSCOntology). For the maintenance phase, Zhong et al. [26] focused on the environmental monitoring, and developed four ontologies to represent building information, sensor information, and regulatory information based on national standards and design requirements, respectively.

In the semantic web technology, ontology technology provides a formal representation of a shared conceptualization [27]. Ontology technology is used to represent concepts and relationships in a specific domain by using formalized languages. Among the relevant languages, OWL is the most widely used one, because it provides more powerful semantic expressiveness and stronger inferential capabilities other languages like XML, RDF, and RDF Schema (RDF-S) [20]. In OWL, OWL classes represent concepts and OWL properties represent relationships [28]. For example, “CbqWall” is an OWL class defined to represent the concept “wall”, and an OWL property “RelatingMaterial” represents the relationship between building components (such as walls) and material (such as concrete). Besides, a set of data is organized as a series of S-P-O (Subject - Predicate - Object, henceforth S-P-O) triples, in which an OWL property links two OWL classes, as shown in Fig. 2.

In Fig. 2, the first OWL class is the domain of the OWL property, and the second OWL class is the range of predicate. Moreover, in the field of ontology technology, several languages have been developed to express rules, in which SWRL is more often used together with OWL.

2.3. Linked building data

The W3C - LBD community group and other researchers developed ontologies to describe building information in the context of linked data. Among these ontologies, ifcOWL is one of the most influential ones, which provides an OWL representation of the IFC standard. The ifcOWL has been recommended by the buildingSMART International [29] and it can be queried by using a structured language, i.e., SPARQL (SPARQL Protocol and RDF Query Language). However, for the task of checking the quality of BIM models, the existing ontologies, such as ifcOWL, are too large in scale and too complex in structure. Thus, it is difficult to reuse the existing ontologies directly.

In addition, Information Delivery Specification (IDS) is a computer-interpretable XML standard developed by the buildingSMART International for defining IFC-based information delivery specifications to facilitate accurate and efficient information exchange. As a tool to

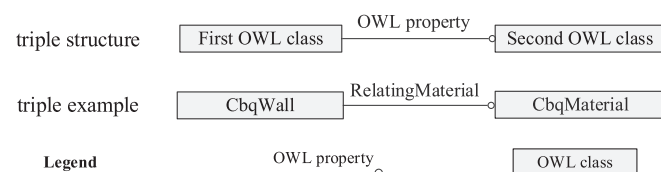


Fig. 2. Structure and example of S-P-O triple in OWL [22].

describe IFC information, the application of ontology such as ifcOWL is complementary to the information transfer process defined by IDS. IDS is concerned not only with the delivery of information, but also with the quality of information. It requires that the information in the BIM model must meet certain standards and specifications to ensure the accuracy and consistency of the information. Kremer et al. [30] discussed about extending the IDS format to model exchange in the building engineering industry to integrate more quality checking resources. They proposed a new extension to the IDS format that enables geometric, topological, and semantic checks to be performed during model checking. They developed a model checker and applied and validated it in the Moodle Model Check project, where the model checker took an IDS file and an IFC model data file as input and returned the results as a BIM Collaboration Format (BCF) file.

2.4. Research gap

According to the above-mentioned review of existing researches, it is concluded that some researches have been carried out regarding only specific aspects of quality checking for BIM models, and existing approaches for automatic compliance check have mainly based on ontology technologies. However, compared with the current research aim, the following research gaps still exist.

First, no research has represented a real standard for automatic quality checking of BIM models. Since existing approaches for the automatic checking BIM models were limited to specific aspects, and a real standard need to be investigated at first to solve the problem as a whole.

Second, no research has proposed a systematic approach for the quality checking of BIM models against real standards. As far as the ontology related to BIM is concerned, the existing ontologies are too large in scale and too complex in structure. Therefore, a systematic approach needs to be developed before the corresponding system is developed because the problem is not only very comprehensive but also very complex.

3. Framework of approach

Although the four-stage process proposed by Eastman et al. has provided a general framework for automatic compliance checking approaches, its customization is needed for such a complex problem as the automatic compliance checking for BIM models against quality standards.

This paper proposes an automatic compliance checking approach for IFC models against quality standards based on ontology technology. As the customization of the four-stage process, the workflow of the approach is given in Fig. 3, where ontology technology is used for the representation of the textual contents of quality standards, as implemented in Steps 1 and 2. Then, the required data is prepared in Step 3, in which the subset of an IFC model is converted into an ontology instance. Further, a rule engine is used to execute checking rules automatically, as implemented in Step 4.

Step 1. Analyze the contents of quality standards for BIM models.

To computerize the contents of the quality standard for BIM models, it is a prerequisite to analyze the concepts and relationships involved in the standard. Therefore, the first step in the workflow is to analyze the standard to extract the concepts and relationships.

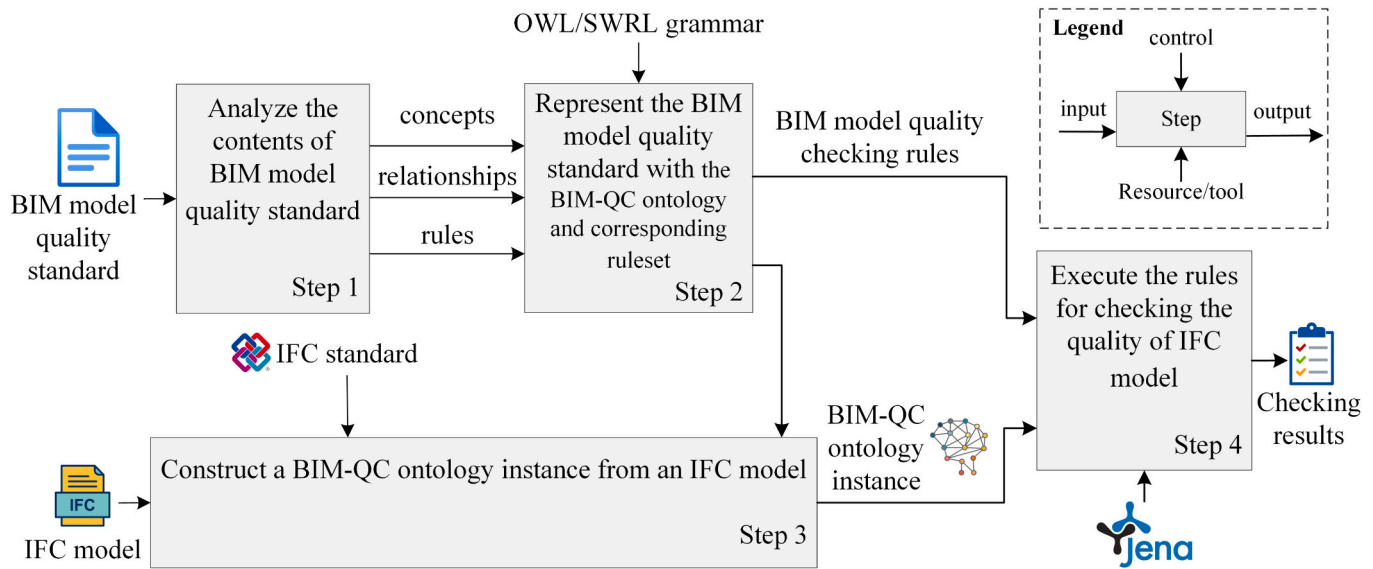


Fig. 3. Overall workflow of proposed approach.

Step 2. Represent the quality standard for BIM models with ontology technology.

The extracted concepts, relationships and checking rules are then represented by using ontology technology. In the process, specifically, OWL classes are used for representing concepts, and OWL properties are used for representing relationships. Then, SWRL is applied to represent the checking rules, because SWRL is tightly integrated with OWL and the atoms of the checking rules in SWRL are OWL classes or properties. The use of SWRL along with OWL computerizes the contents of quality standard into the BIM-QC ontology and a corresponding ruleset.

Step 3. Construct an instance of the BIM-QC ontology from an IFC model.

BIM-QC ontology indicates the required information for the automatic compliance checking for IFC models against quality standards. In this step, the required information is extracted from the IFC model to form a subset of model data. Because the checking rules in SWRL can only be executed on corresponding ontology instance, the subset of model data need to be converted into a BIM-QC ontology instance.

Step 4. Execute the checking rules for the quality of IFC models.

With the support of rule engines, such as Jena [31] and OWL API [32], the checking rules in SWRL are executed automatically to check the compliance for the IFC model against quality standards.

Note that **Steps 1** and **2** are executed only once, while **Step 3** and **4** need to be done for each IFC model to be checked. In the following, this paper only focuses on the **Steps 1 to 3** in [Sections 4 to 6](#) respectively, corresponding to the above-mentioned research questions, because **Step 4** is implemented by using existing tools in this study.

4. Analysis of quality standard for BIM models

In this section, the contents of the quality standard for BIM models are analyzed to extract the concepts, relationships, and checking rules by examining the standard carefully.

4.1. Analyze concepts in quality standards for BIM models

The quality standard for BIM models specifies the requirement on the data completeness, correctness and consistency of BIM models. Most quality standards for BIM models, including the Chinese standard, are composed of a series of clauses. These clauses indicate the basic and specific requirements on BIM models. The contents of the Chinese standard consist of 5 chapters. Chapter 1 introduces the purpose of the

standard and Chapter 2 lists the terms used in the standard. Then a series of requirements about BIM model quality are contained, including basic quality requirements for BIM model (Chapter 3), specific quality requirements for BIM model units (Chapter 4) and other requirements for deliveries related with BIM models (Chapter 5), such as 2D drawings. Since this research focuses on BIM models, Chapters 3 and 4 are the main contents to be computerized in this paper, which contain 15 clauses and 21 clauses, respectively.

By analyzing the semantics of the clauses in the Chinese standard one by one, it is observed that each clause represents one or more checking rules on the following two categories.

- (1) The feature of the objects in the BIM model; and
- (2) The relationship between the objects in the BIM model.

Here, the object refers to any perceivable things mentioned in the BIM model, such as walls and slabs. The feature refers to the characteristics of the objects, such as color and material.

Furthermore, from the perspective of quality dimensions, the checking rules contained in the clauses of quality standards for BIM models are classified into categories of completeness, correctness and consistency, as shown in [Table 1](#).

Based on the analysis and classification of the checking rules in the Chinese standard, the concepts in it are generally divided into two types, i.e., object and feature. Namely, object and feature are extracted as the top-level concepts and other concepts are derived from them.

By reviewing the clauses in the Chinese standard one by one, the concepts derived from object are further divided into four categories, i.e., project, system, component, and part. Each category can be further classified into more detailed sub-categories.

By reviewing the characteristics scattered in the clauses of the Chinese standard, the concepts derived for features are classified into four types of features, i.e., geometry features, space features, surface features and attribute features.

Finally, 179 concepts are extracted and generalized from the Chinese standard. Several examples are provided in [Fig. 4](#) and Appendix A.

4.2. Analyze relationships in quality standards for BIM models

Considering that only two top-level concepts are extracted and classified from the Chinese standard, logically and obviously, three types of potential relationships can exist between these concepts, i.e., the

Table 1

Classification of checking rules in quality standard for BIM models.

No.	Classification criteria	Description	Example checking rules
1	I Quality dimension: Completeness		
2	I-A Complete feature	BIM objects should have all prescribed features	The project should have an associated coordinate system
3	I-B Complete relationship	BIM objects should have all prescribed relationships	The components of the peripheral system should indicate connection relationship
4	II Quality dimension: Correctness		
5	II-A Correct features	The values of features should be correct	The components of fire protection system should be red
6	II-B Correct relationship	The relationship between BIM object should be correct	The components of different materials should not overlap each other
7	III Quality dimension: Consistency		
8	III-A Consistent feature	The values of the same feature in different places should be consistent	The color of any component should be consistent with that of its system
9	III-B Consistent relationship	The same relationships represented in different places should be consistent	The project to which the component belongs should be consistent with the project to which the system containing it belongs

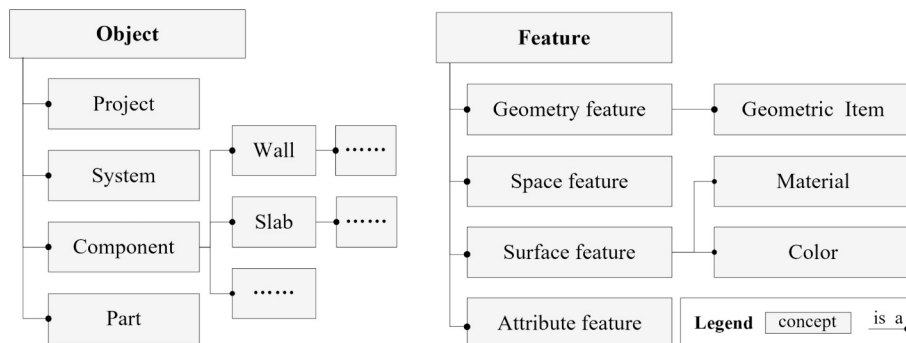
relationships within features, the relationships within objects and the relationships between objects and features. Among the three types, the relationship within features, such as that between color and material, has no physical meaning in practice, and it also does not exist as a constraint in the Chinese standard. Thus, only the latter two types of relationships are extracted and classified in Table 2 by reviewing the clauses of the Chinese standard one by one.

4.3. Analyze checking rules in quality standards for BIM models

Based on the concept and relationship, the checking rules for the quality of BIM models in the Chinese standard are extracted and decomposed. By analyzing the semantics of the checking rules, it is found that the checking rules for the quality of BIM models have the same composition and structure.

By analyzing the structure, it is found that these checking rules can be decomposed into the following 4 compositions, the examples of which are shown in Table 3.

- (1) The type of the object to be checked. For example, the type of the object in Checking Rule 1 of Table 3 is “project”.
- (2) The scoping conditions for the object to be checked (can be default). Some checking rules are only for a certain type of object.

**Fig. 4.** Examples of concepts in quality standards for BIM models.

For example, Checking Rule 3 in Table 3 only constrain the components in fire protection system. Thus, in these cases, one or more scoping conditions of the objects to be checked are declared in checking rules.

- (3) The feature of the objects or the relationship between the objects to be checked. For example, in Table 3, the feature to be checked of Checking Rule 3 is color.
- (4) The requirement. The checking rule needs to indicate the explicit requirement on the completeness, consistency or correctness. For completeness checking rules, the requirement is “should have”, such as Checking Rule 1 in Table 3. For consistency checking rules, the requirement is “should be consistent”, such as Checking Rule 2 in Table 3. For correctness checking rules, the requirement

Table 2

Relationships in quality standards for BIM models and examples.

No.	Relationship	Description	Examples
1	RelatingFeature	The relationship between objects and their features	The relationship between a wall and concrete material
2	ConnectWith	Connection between two components	The connection between a water pipe and an elbow
3	Contain	The containing relationship between low-level and high-level objects	The dependency between a rebar and a reinforced concrete beam
4	IntersectWith	The intersecting relationship between two components	The spatial intersection between a concrete beam and a column

Table 3

Example checking rules in Chinese standard and compositions of checking rules.

The Checking Rules in the clauses of the Chinese standard	Composition of checking rules			
	Type	Scoping conditions	Feature/relationship	Requirement
1 The project should have an associated coordinate system	Project	None	Coordinate system	Should have
2 The color of component should be consistent with that of its system	Component	None	Color	Should be consistent
3 The components in fire protection system should be red	Component	Fire protection system	Color	Should be red

is the prescribed value of the feature or relationship. For example, the required prescribed color is red in Checking Rule 3 in Table 3.

5. Representation of quality standard by using ontology technology

In this section, the Chinese standard are represented by using ontology technology. Specifically, the concepts are represented with OWL classes as stated in Section 5.1, and the relationships are represented with OWL properties as stated in Section 5.2, forming an ontology called BIM-QC ontology. Based on the OWL classes and properties in the BIM-QC ontology, the checking rules for the quality of BIM models are computerized into a ruleset in SWRL in Section 5.3.

5.1. Represent concepts with OWL classes

Each concept in the Chinese standard is represented as an OWL class. Correspondingly, the individual of OWL classes (hereafter referred to as “OWL individual”) can be defined in the ontology instance. For the two top-level concepts in the quality standards for BIM models, corresponding base classes are defined, i.e., CbqQualityCheckObject for objects and CbqFeature for features. More detailed concepts are represented by using subclasses. For example, four subclasses of CbqFeature are defined as CbqGeometryFeature, CbqSpaceFeature, CbqSurfaceFeature, and CbqAttributeFeature, to represent different types of features. Several examples are provided in Fig. 5 and the complete list of OWL classes and properties is given in Appendix A.

5.2. Represent relationships with OWL properties

The relationships are represented by OWL properties. Thus, each relationship in quality standard for BIM models is represented as an OWL property in the ontology. Respectively, the four relationships in the Chinese standard are represented as four types of OWL properties, i.e., RelatingFeature, ConnectWith, Contains and IntersectWith, corresponding to Property I, II, III, IV in Table 4. In addition, sub properties are declared via the predefined OWL property subPropertyOf. For example, in Table 4, RelatingColor (I.A) is a sub property of RelatingFeature, which is used for representing the color of a BIM object.

It deserves to emphasize that both OWL classes and OWL properties constitute the ontology called BIM-QC ontology, which lays the foundation of the representation of the quality standard.

5.3. Represent checking rules in SWRL

Normally, a checking rule in SWRL is composed of two parts, i.e., the conditions and the consequence [33]. Both the conditions and the consequence consist of one or more atoms. According to the grammar, an atom in SWRL is of one of the following four forms, where ?x and ?y are individuals to represent a certain OWL class or any basic data type

[33].

- (1) Class (?x) represents a certain class of OWL individuals.
- (2) Property (?x,?y) represents the relationship between an OWL individual and another or a value of basic data type.
- (3) sameAs(?x,?y) indicates two OWL individuals of the same concept.
- (4) differentFrom(?x,?y) indicates two OWL individuals of different concepts.

In a checking rule in SWRL, the atoms for conditions are connected with a symbol “^”, and then the symbol “→” means the action to get a consequence according to these conditions. Based on the decomposition of the rules for checking the quality of BIM models and the grammar of SWRL, an SWRL-based representation process of the checking rules is proposed. In the process, the four compositions are represented sequentially by using SWRL atoms, i.e., the type of the object to be checked, the scoping conditions for the object to be checked (can be default), the feature of the object or the relationship between the objects to be checked, and the requirement. Then, the conclusion of compliance checking is drawn by using an atom.

More specially, the checking rule representation process consists of the three stages, i.e., **Declare**, **Judge** and **Conclude**. Among them, the stages **Declare** and **Judge** construct the atoms of conditions in a checking rule in SWRL. Then the stage **Conclude** derives the consequence of the checking rule in SWRL. The implementation of each stage is illustrated in Fig. 6, where a checking rule is computerized as an example. The original text of the checking rule in the Chinese standard is “the color of the component in fire protection system should be red”.

In the stage **Declare**, the first three compositions of a checking rule are declared through a series of atoms. Firstly, the type of the object to be checked is declared by an atom of Class (?x). In the example as shown in Fig. 6, the type of object to be checked is a component and the corresponding atom is CbqComponent (?x). Secondly, the declaration for scoping conditions of the object is completed by several atoms of Property (?x,?y). Each scoping condition consists of two parts, i.e., the feature or relationship used for defining the scope and its values. Accordingly, each scoping condition is generally declared by using a pair of atoms. In the example in Fig. 6, the checking rule only applies to the components in fire protection system, thus the relationships used for defining the scoping condition is “Contains” and the value is “fire protection system”, so the pair of atoms to declare the scoping conditions are Contains (?y,?x) and CbqFireProtectionSystem (?y). Thirdly, the feature or the relationship to be checked is declared by an atom of Property (?x,?y). In the example in Fig. 6, the feature to be checked is color, thus corresponding atom is RelatingColor (?x,?z).

Then, in the stage **Judge**, the next atom judges whether the feature or relationship to be checked satisfies the requirements in the checking rule. According to the grammar of SWRL, the atom that undertakes the judgement differs for the checking rules of different quality dimensions.

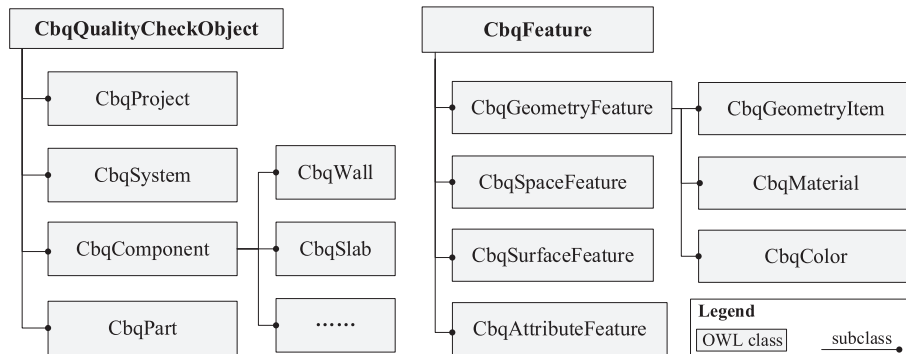


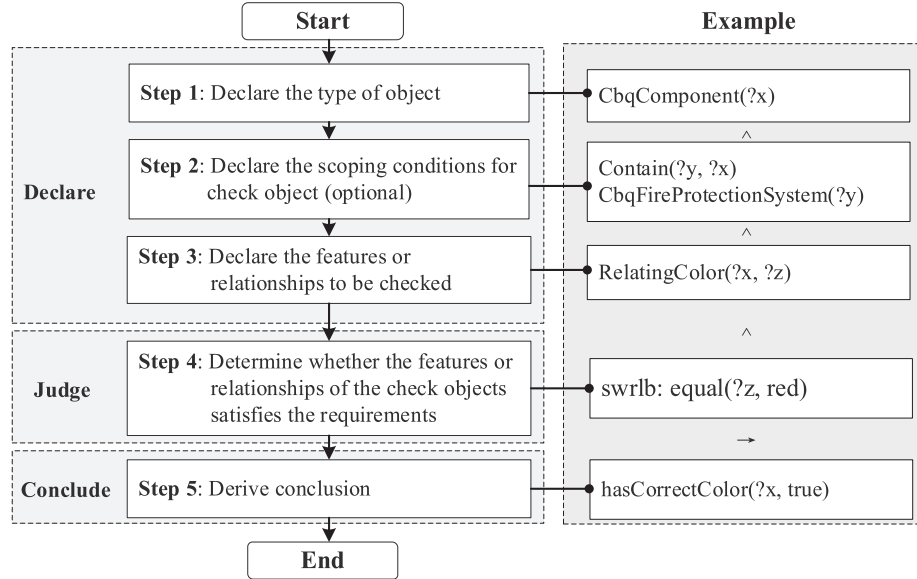
Fig. 5. OWL-based representation for concepts in quality standards for BIM models.

Table 4

Examples of definition of OWL properties.

No.	Property		Domain	Range	Description	Property Type
1	I	Relating Feature	OQCO	CbqFeature	Feature of BIM object	object property
2	I.A	RelatingColor	OC	CbqColor	Color of BIM object	object property
3	I.B	RelatingIdentifier	OQCO	String	Identifier of BIM object	datatype property
4	II	ConnectWith	OC	OC	Connection relationship between BIM component	object property
5	III	Contains	OQCO	OQCO	Containment relationship between BIM component	object property
6	IV	IntersectWith	OC	OC	Intersecting relationship between BIM component	object property

Note. OQCO refers to CbqQualityCheckingObject, OC refers to CbqComponent.

**Fig. 6.** Process to construct checking rules in SWRL and example.

For the completeness checking rules, the requirement is “should have”, thus the corresponding atom for judgement is in the form of $(Property > 0) (?x, ?y)$, which means the value of the feature or the relationship is not empty. For consistency checking rules, atom can be constructed by the property $sameAs (?x, ?y)$. For the correctness checking rules, it can be determined whether the value of the feature is equal to the value prescribed in the standard. The example as shown in Fig. 4 is a checking rule for correctness and the atom for judgement is $equal (?z, red)$.

Finally, in the stage **Conclude**, an atom $CorrectColorRule(?x, true)$ is constructed to conclude that the object satisfies this checking rule. The converted checking rule is given in Listing 1.

By applying the process, the checking rules in the Chinese standard are represented into a computer readable format and a ruleset composed of 23 checking rules in SWRL is established, as shown in Appendix B. The 23 checking rules corresponds 25 clauses in the Chinese standard, which constitute 83% among the total 30 relevant clauses in the standard.

Listing 1. Example of checking rule in SWRL.

```

CbqComponent(?x) ^ // declare the type of object to be checked
Contains(?y, ?x) // declare the relationship of scoping condition
^ CbqFireProtectionSystem(?y) ^ // declare the value of scoping
condition
RelatingColor (?x, ?z) ^ // declare the feature to be check
swrlb: equal(?z, red) // judges whether the feature to be checked
satisfies the requirements
→ hasCorrectColor(?x, true) // conclude the checking result
  
```

6. Automatic checking of BIM models against quality standard

In this approach, for an IFC model, a BIM ontology instance needs to

be constructed to execute the checking rules in SWRL correspondingly. Considering that an ontology instance consists of OWL individuals and OWL property, Sections 6.1 and 6.2 address two critical issues, i.e., the acquisition of OWL individuals and that of OWL properties from an IFC model, respectively.

6.1. Acquire OWL individual from IFC model

In the BIM-QC ontology, CbqQualityCheckObject and CbqFeature are defined as base OWL class to represent the top-level concepts, i.e., object and feature. To construct the OWL individuals of the BIM-QC ontology instance corresponding to an IFC model, it is essential to clarify how these concepts are represented in the IFC standard and then build a map from IFC entities to OWL classes in the BIM-QC ontology. This issue has been resolved by carefully examining the contents of the IFC standards.

The OWL class CbqQualityCheckObject represents any perceivable things in the BIM model, such as walls and slabs, which are expressed by using IfcObjectDefinition entity and its derived entities in the IFC standard. CbqQualityCheckObject has four subclasses, i.e., CbqProject, CbqSystem, CbqComponent and CbqPart. Correspondingly, the four derived entities from IfcObjectDefinition entity, i.e., IfcProject, IfcSystem, IfcElement and IfcElementComponent, represent these concepts. More details and examples of the mapping from IfcObjectDefinition entity to CbqQualityCheckObject class are given in Table 5 and Appendix A.

The OWL class CbqFeature represents the characteristics of objects, such as color and material. In the IFC standard, these concepts are expressed by using the entities in resource layer, in which various supporting entities are defined to describe the building information, such as

Table 5

Examples of mapping from IFC entity to OWL class for objects.

No.	Concept	IFC entity	OWL class
1	I	Any perceivable things	CbqQualityCheckObject
2	I.A	Project	CbqProject
3	I-B	System	CbqSystem
4	I-B.1	Architectural and structural system	CbqBuildingSystem
5	I-B.2	Distribution system	CbqDistributionSystem
6	I-C	Tangible physical product, such as beam and pipe	CbqComponent
7	I.C.1	Tangible physical product in architectural system	CbqBuildingComponent
8	I.C.1.1	Wall	CbqWall
9	I.C.1.1	Column	CbqColumn
10	I.C.2	Tangible physical product in distribution system	CbqDistributionComponent
11	I-D	Minor items in component, such as rebar in a beam	CbqPart

geometric item, material, and property. More details of the mapping from IFC entities in resource layer to CbqFeature class are given in Table 6 and Appendix A.

According to the mapping, for an IFC model, the OWL individuals of BIM-QC ontology instance can be constructed directly. For example, if an IFC model contains an IfcWall occurrence, an individual of CbqWall class is created in the BIM-QC ontology instance correspondingly.

6.2. Acquire OWL property from IFC model

On the basis of OWL individuals, an OWL property needs to be instantiated to represent the relationship between the OWL individuals and further form a complete BIM-QC ontology instance. Also, to construct an OWL property corresponding to an IFC model, it is essential to clarify how these relationships are represented in the IFC standard. This issue has also been resolved by carefully examining the contents of the IFC standards.

As stated in Section 5.2, the BIM-QC ontology contains four types of OWL properties representing the relationships in the Chinese standard, i. e., RelatingFeature, Contains, IntersectWith, and ConnectWith. As a result of a thorough investigation on the IFC standard, these relationships are expressed as follows.

- (1) Regarding the OWL property **RelatingFeature**, Fig. 7 and Fig. 8 indicate the expression of the relationships between objects and features in the IFC standard.

Fig. 7 expresses the three types of features in the IFC standard, i.e., space feature, attribute feature and geometry feature. **Part A** expresses the association between IFC objects and the space features. IfcObject-Placement is associated with IfcProduct directly to represent the coordinate system information. Meanwhile, the IfcSpace is linked with the IfcElement through the relationship entity IfcRelContainedInSpatial-Structure. **Part B** indicates the expression of the attribute features in the IFC standard. The attribute feature is expressed by using IfcProperty, which is associated with the IFC object through the relationship entity IfcRelDefinesByProperties. As **Part C** indicates, the geometry feature is expressed by using IfcGeometricRepresentationItem, whose derived

entities define various types of geometric representations such as point, curve and solid. IfcGeometricRepresentationItem is linked with the IFC object with the two entities, i.e., IfcProductRepresentation and IfcShapeRepresentation.

Fig. 8 indicates the expression of surface feature in the IFC standard, including material and color. **Part A** expresses the association between IFC objects and material. Material feature is expressed by using IfcMaterial, which is associated with the IFC object through the relationship entity IfcRelAssociatesMaterial. **Part B** expresses the color features in the IFC standard, which is expressed by using IfcColourRgb and associated with the IFC object through a series of intermediate entities. Note that the expressions of material and color features are varied in the IFC standard due to the complexity and flexibility. Fig. 8 shows one of them and the Literature [34] proposed a method to obtain the various expressions.

- (2) Regarding the OWL property **Contains**, Part A and B in Fig. 9 express the containment relationships between objects.

Part A indicates the expression of containment relationship between components and parts, such as rebar in a beam. In the IFC standard, the relationship entity IfcRelAggregates is used to express the decomposition relationships between a component and its part. **Part B** expresses the containment relationship between a system and a component. In the IFC standard, the relationship entity IfcRelAssignsToGroup is used to assign a component to a system and represent the containment relationship between them.

- (3) Regarding the OWL property **IntersectWith**, Part C in Fig. 9 expresses the intersecting relationships between objects on spatial topology. In the IFC standard, the relationship entity IfcRelAggregates expresses the intersecting relationships.
- (4) Regarding the OWL property **ConnectWith**, Fig. 10 express the connection relationships between objects, in which Part A and Part B indicate the two types of expressions.

Especially, **Part A** expresses the connection relationship between the

Table 6

Examples of mapping from IFC entity to OWL class for features.

No.	Concept	IFC entity	OWL class
1	I Geometry feature		
2	I.A	Geometric item	CbqGeometryItem
3	II Space feature		
4	II.A	Coordinate system	CbqCoordinateSystem
5	II.B	Occupied space	CbqSpaceOccupancy
6	III Surface feature		
7	III.A	Material	CbqMaterial
8	III.B	Color	CbqColor
9	IV Attribute feature		
10	IV.A	Attached attribute	CbqProperty

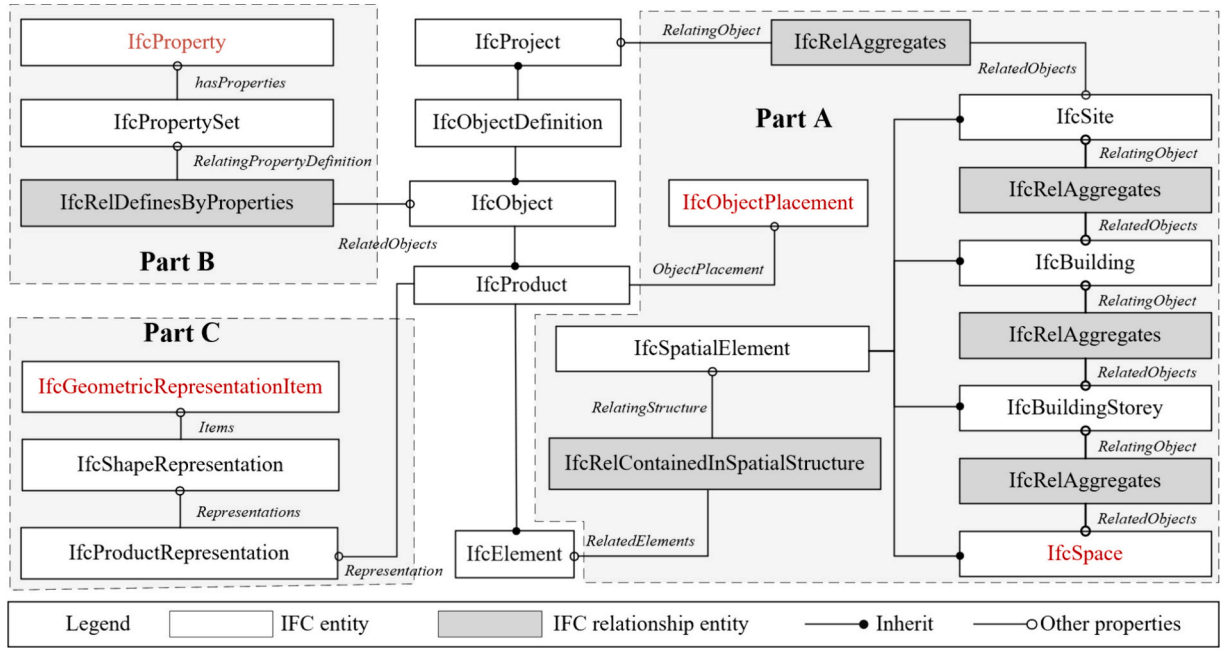


Fig. 7. Expression of space feature, attribute feature and geometry feature in IFC standard.

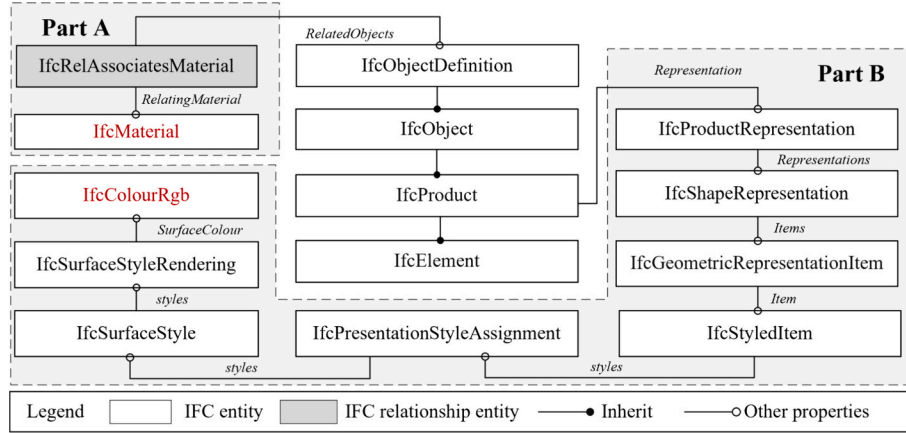


Fig. 8. Expression of space surface feature in IFC standard.

components in the architectural or structural system by using a relationship entity *IfcRelConnectsElements*, such as walls. **Part B** illustrates the expression of connection relationship between the components in the MEP (Mechanical, Electrical and Plumbing) system by using relationship entities *IfcRelConnectsPortToElement* and *IfcRelConnectsPorts*.

After obtaining the expressions of the required relationships in the IFC standard, the corresponding OWL properties can be created in the BIM-QC ontology instance for an IFC model by the following three steps due to obvious reasons.

- (1) Obtain an OWL individual and corresponding IFC object occurrence. For example, an OWL individual *CbqWall_3758* in the BIM-QC ontology instance and corresponding object occurrence in the IFC model are obtained.
- (2) Find the value in the IFC model and corresponding OWL individual for a type of relationship. For example, if the target relationship is *RelatingMaterial*, the occurrence of *IfcMaterial* which is associated with *IfcWall_3758* can be found directly according to Part B of Fig. 8 from the IFC model, such as *IfcMaterial_4987*

with the label of “concrete”. Then corresponding OWL individual *CbqMaterial_4987* is obtained accordingly.

- (3) Create the assertion to declare OWL property between OWL individuals. For example, an assertion to declare the *RelatingMaterial* relationship between *CbqWall_3758* and *CbqMaterial_4987* is created in the BIM-QC ontology instance.

After creating all the OWL individuals and OWL properties, a BIM-QC ontology instance is thus obtained from an IFC model. Then the checking rules in SWRL can be executed on it by using a rule engine to check the compliance for the IFC model against quality standards automatically.

7. Prototype system and case study

To validate the feasibility and effectiveness of the proposed approach, a prototype system has been developed. The structure of the prototype system and the open-source tools used in it are shown in Fig. 11. In the process, Protégé [35] is used to edit the ontology in OWL and checking rules in SWRL to establish the BIM-QC ontology and the

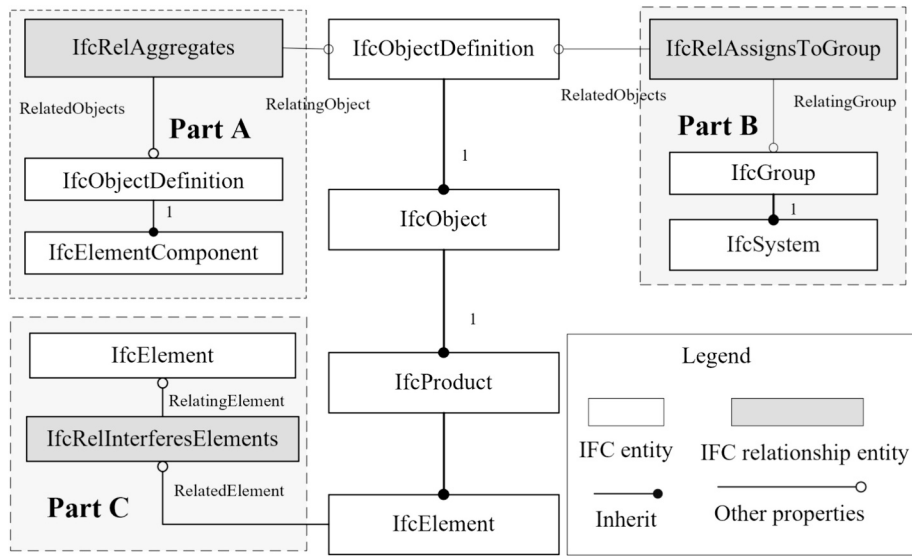


Fig. 9. Expression of relationships **Contains** and **IntersectWith** in IFC standard.

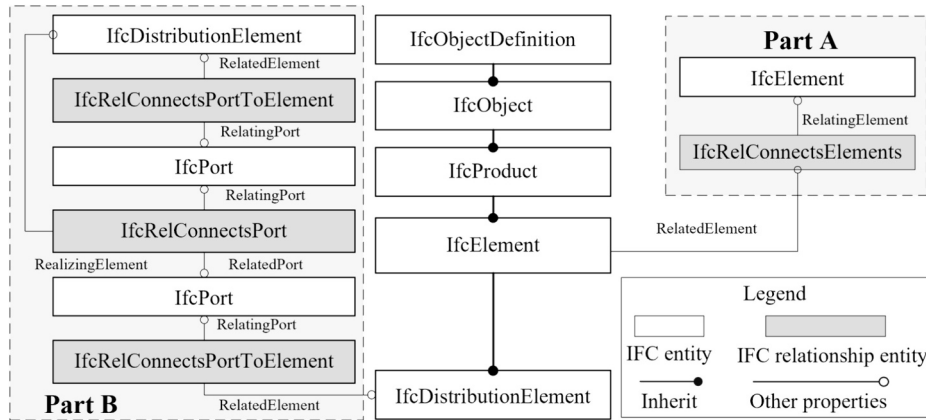


Fig. 10. Expression of relationship **ConnectWith** in IFC standard.

ruleset. Jena and OWL API rule engine [31] are used to execute the checking rules in SWRL to check the compliance for BIM model against quality standards, and BIMserver [36] is used to visualize the IFC model and checking result. Both the latter two components are integrated into the prototype system. As a case study, the proposed approach is tested by using an IFC model for an office building project [37]. Fig. 12 shows the

interface of the prototype system and the IFC model for test.

The actual maturity of the prototype system needs to be discussed in detail before verification. First of all, 38 requirements from the total 48 requirements in the body of the two standards are computerized as checking rules, mainly including rules about color, naming, model fineness and model content, but not including the requirements for the

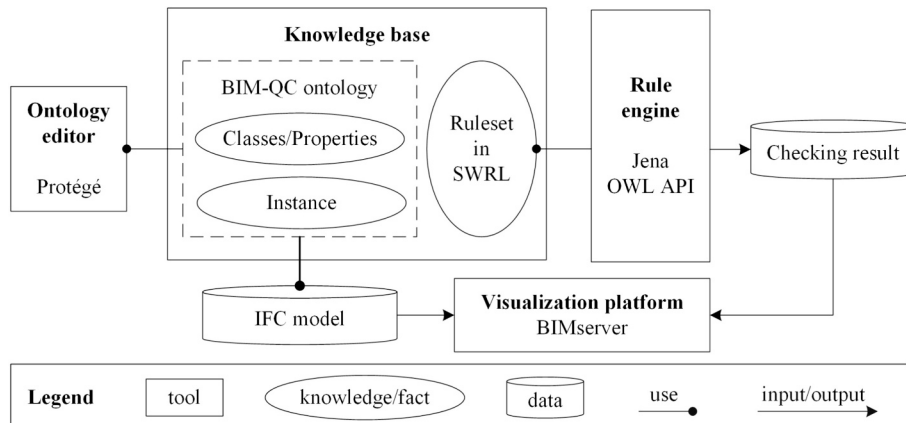


Fig. 11. Structure of prototype system.

levels of geometric detail and information detail of BIM model components in the appendix of the standards. Second, the prototype system can check the quality of common components such as walls, panels, doors, windows etc. at present. Obviously, these limitations in the prototype system do not affect the realization of the research objective of the research.

In the prototype system, the checking rules in SWRL were executed to check the quality of a large and complex IFC model of an office building [38]. The IFC model size is 77.70 MB and contains a total of 5396 components. Then, the checking results of the different types of component instances were exported and visualized in BIMserver as shown in Fig. 13.

In Fig. 13, users can choose to view the checking results for any instance of different kinds of components and the checking results are divided into the three types, i.e., Pass, Fail and Unknown. Pass and Fail indicate whether the current IFC model satisfies this checking rule, respectively. Unknown means that the information in the current IFC model is insufficient to support the judgement for checking rules. For example, Checking Rule 9 in Fig. 13 is “Component should have the consistent color with its system”, but the result of Checking Rule 5 indicates that the wall is not contained in any system. Therefore, the judgement for Checking Rule 9 cannot be made. Thus, the proposed approach is validated.

It is worth mentioning that in terms of scalability, the prototype system has obvious advantages compared with existing similar systems that Lee et al. sum up [35]. In this study, OWL ontology and SWRL rule base are used to increase and expand the functions. Users do not need to master much complex programming knowledge, except understanding the OWL language and SWRL language syntax when they expand the checking rules.

8. Conclusions

This paper proposed an ontology-based method for the automatic compliance checking for BIM models against quality standards. The major findings and contributions to knowledge of this paper are concluded as follows.

- (1) The key concepts and relationships of the Chinese standards for quality of BIM model could be expressed by using ontology technology, and most of the checking rules for BIM model quality could be represented as well.
- (2) A process including representing the standard for quality of BIM model by using OWL and SWRL, establishing the mapping between the ontology of the standard and IFC model, and checking the quality of BIM model by extracting the relevant information based on the mapping and IFC model was established, which could be used for the automatic compliance checking for the quality of BIM models.
- (3) A prototype system for automatic compliance checking of BIM model quality was developed, which demonstrates the feasibility and effectiveness of the proposed method.

The method is expected to be usable not only for other Chinese standards for the quality of BIM model, but can be used for the similar standards of other countries. Overall, this research contributes not only to the development of automatic compliance checking systems for the quality of BIM models against real standards, but also to the promotion of BIM technology.

The proposed approach lays a sound foundation for developing an automatic compliance checking system for the quality of BIM models to improve efficiency, reduce errors in BIM model and save human resources. The strength of the approach is that it provides a systematic and easy-to-adopt approach to computerize two real standards for compliance checking for the quality of BIM models based on ontology technology. This research contributes to the development of automatic compliance checking systems for the quality of BIM models against real standards and the promotion of BIM technology.

This study still has limitations. The approach could not computerize all the checking rules for the BIM model quality. For example, regarding the checking rule “The information of model units should be added and deleted according to application requirements (Clause 3.1.3 in the Chinese standard)”, it cannot be represented by using ontology technology, because the term “application requirements” in the clause depends on particular cases. Up to now, 38 requirements from the total 48 requirements related to BIM model quality in the two Chinese standards

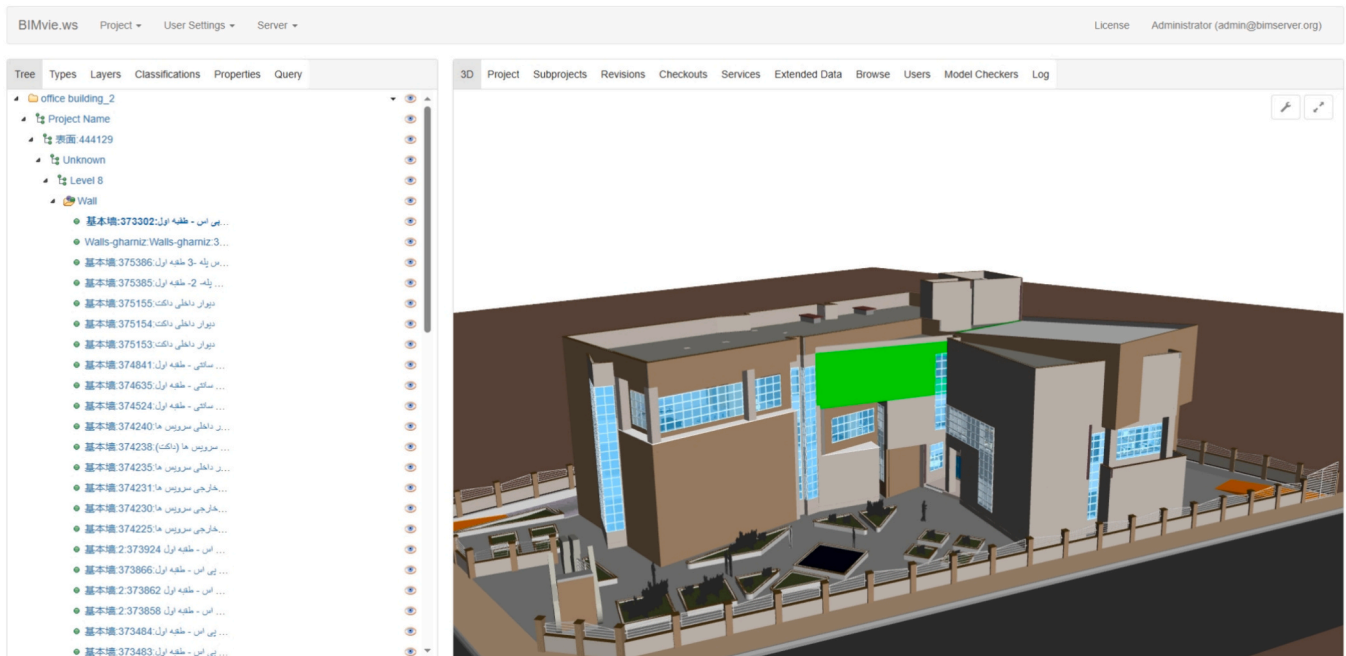


Fig. 12. IFC model of office building BIM model in BIMserver.

The screenshot shows the BIMserver interface with the 'Extended Data' tab selected. The top navigation bar includes links for 3D, Project, Subprojects, Revisions, Checkouts, Services, Extended Data, Browse, Users, Model Checkers, and Log. Below the navigation bar, there's a section for 'Showing extended data for' with a dropdown menu set to 'Revision 1 (Last)' and a button 'Add extended data'. A 'back' link is also present.

The main content area displays the following metadata for 'IFC Validator':

Title	IFC Validator
User	Administrator
Schema	GEOMETRY_GENERATION_REPORT_JSON_1_1
Size	22.92 kB
Mime type	text/html
Download	Download
Added	29-06-2024 10:26

Below the metadata, there's a section titled 'IFC Validator (HTML)'. It includes two dropdown menus: 'Types of component' (set to 'Wall') and 'Instances of chosen component' (set to 'Wall_132331'). A blue button 'View checking results' is located below these dropdowns.

The 'Results for Wall_132331' section contains a table with 10 checking rules:

Rule	Index	Actual value	Required value
✓ Checking Rule 1 Component should have color information.	None	2	>0
✓ Checking Rule 2 Component should have at least an associated coordinate system.	None	1	>0
✓ Checking Rule 3 Component should be associated with at least one location point.	None	1	≥1
✓ Checking Rule 4 Component should have material information.	None	3	>0
⚠ Checking Rule 5 Component should be contained in at least one system.	None	0	>0
✓ Checking Rule 6 Component with different materials should not overlap with each other.	None		
⚠ Checking Rule 7 The component should have the correct level of information detail.	None	N2	N3
✓ Checking Rule 8 The components of the building peripheral system should indicate direct connection relationships.	None		
ⓘ Checking Rule 9 The color of the component should be consistent with the color of its system.	None		
✓ Checking Rule 10 Components should have a consistent coordinate system with the project.	None	consistent	consistent

Fig. 13. Checking results in interface of BIMserver.

have not been computerized as checking rules, accounting for about 20%. Besides, the requirements on different levels of geometric detail and information detail for different types of BIM model components in the appendices of the two standards have not been computerized. In addition, the processing of IFC model in the prototype system currently can only extract information of common component types such as walls, slabs, doors, windows etc. for the automatic generation of BIM-QC ontology instance.

In future work, we will study methods to improve the computerized representation rate of the requirements of the standard, and further realize the checking of the level of geometric detail and information detail of the BIM model components. In addition, we will integrate a more mature IFC model parser into the prototype system to improve the functionality of the system to enhance the ability of the prototype system.

The BIM-QC ontology and the corresponding ruleset, the source code of the prototype system, and related documents have been published in

GitHub [38] for further applications by the interested readers of this paper for free.

CRediT authorship contribution statement

Zhiliang Ma: Writing – review & editing, Writing – original draft, Methodology. **Honggang Zhu:** Writing – review & editing, Writing – original draft. **Xinglei Xiang:** Writing – original draft, Methodology. **Ziga Turk:** Writing – review & editing, Methodology. **Robert Klinc:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Concepts in Chinese standard and corresponding OWL classes/IFC entities**Table A-1**

Concepts and corresponding OWL classes and IFC entities.

Concept	OWL class	IFC entity
1 Model unit	CbqModelUnit	IfcObjectDefinition
1.1 Project	CbqProject	IfcProject
1.2 System	CbqSystem	IfcSystem
1.2.1 Peripheral system	CbqPeripheralSystem	IfcBuildingSystem
1.2.2 Other system	CbqOtherBuildingSystem	
1.2.3 Plumbing system	CbqWaterSystem	IfcDistributionSystem
1.2.4 HVAC system	CbqHvacSystem	
1.2.5 Electric system	CbqElectricSystem	
1.2.6 Intelligent system	CbqIntelligentSystem	IfcDistributionCircuit
1.2.7 Power system	CbqPowerSystem	
1.3 Component	CbqComponent	IfcElement
1.3.1 Building Component	CbqBuildingComponent	IfcBuildingElement
1.3.1.1 Wall	CbqWall	IfcWall
1.3.1.2 Column	CbqArchitecturalColumn	IfcColumn
1.3.1.3 Door	CbqDoor	IfcDoor
1.3.1.4 Window	CbqWindow	IfcWindow
1.3.1.5 Roof	CbqRoof	IfcRoof
1.3.1.6 Curtain wall	CbqCurtainWall	IfcCurtainWall
1.3.1.7 Ceiling	CbqCeiling	IfcCovering
1.3.1.8 Stair	CbqStair	IfcStair
1.3.1.9 Ramp	CbqRamp	IfcRamp
1.3.1.10 Railing	CbqRailing	IfcRailing
1.3.1.11 Apron	CbqApron	IfcBuildingElementProxy
1.3.1.12 Furniture	CbqFurniture	IfcFurnishingElement
1.3.2 Structure component	CbqStructuralComponent	IfcBuildingElement
1.3.2.1 Foundation	CbqFoundation	IfcFooting/IfcPipe
1.3.2.2 Structure wall	CbqStructuralWall	IfcWall
1.3.2.3 Structure Column	CbqStructuralColumn	IfcColumn
1.3.2.4 Beam	CbqBeam	IfcBeam
1.3.2.5 Slab	CbqSlab	IfcSlab
1.3.3 Plumbing Component	CbqWaterComponent	IfcDistributionElement
1.3.3.1 Water Equipment	CbqWaterEquipment	IfcFlowMovingDevice
1.3.3.2 Pool	CbqPool	
1.3.3.3 Tank	CbqTank	IfcTank
1.3.3.4 Water pipe	CbqWaterPipe	IfcPipeSegment
1.3.3.5 Pipe Fitting	CbqPipeFitting	
1.3.3.6 Pipe Accessory	CbqPipeAccessory	IfcPipeFitting
1.3.4 Hvac component	CbqHvacComponent	IfcDistributionElement
1.3.4.1 Hvac equipment	CbqHvacEquipment	IfcEnergyConversionDevice
1.3.4.2 Duct	CbqDuct	IfcDuctSegment
1.3.4.3 Duct fitting	CbqDuctFitting	
1.3.4.4 Duct accessory	CbqDuctAccessory	IfcDuctFitting
1.3.4.5 Duct support	CbqDuctSupport	IfcBuildingElementProxy
1.3.5 Electrical component	CbqElectricalComponent	IfcDistributionElement
1.3.5.1 Electric equipment	CbqElectricalEquipment	IfcElectricFlowStorageDevice
1.3.5.2 Cable tray	CbqCableTray	IfcCableSegment
1.4 Part	CbqPart	IfcElementComponent
2 Feature	CbqFeature	
2.1 Geometry feature	CbqGeometryFeature	
2.1.1 Geometric element	CbqGeometricElement	IfcGeometricRepresentationItem
2.2 Space feature	CbqSpaceFeature	
2.2.1 Location point	CbqLocationPoint	IfcCartesianPoint
2.2.2 Coordinate system	CbqCoordinateSystem	IfcLocalPlacement
2.2.3 Space occupy	CbqSpaceOccupy	IfcSpace
2.3 Surface feature	CbqSurfaceFeature	
2.3.1 Color	CbqColour	IfcColourRgb
2.3.2 Material	CbqMaterial	IfcMaterial
2.4 Attribute	CbqAttributeFeature	IfcProperty

Appendix B. Checking rules for quality BIM models in SWRL

Table B-1

Checking rules for quality BIM models in SWRL.

No.	Checking rule (Original text)	Checking rule in SWRL
1	The system should have an associated coordinate system	<code>CbqSystem(?x)^(RelatingCoordinateSystem≥ 1)(?x) → SystemHasCoordSystem(?x, true)</code>
2	The project should have an associated coordinate system	<code>CbqProject(?x)^(RelatingCoordinateSystem≥ 1)(?x) → ProjectHasCoordSystem(?x, true)</code>
3	The coordinate system associated with the system should be consistent with that associated with the project	<code>CbqProject(?x)^(RelatingCoordinateSystem(?x,y)^(CbqCoordinateSystem(?y)^(CbqProject(?z)^(Contains(?x,z)^(RelatingCoordinateSystem(?z,y) → SystemHasConsistentCoordinateSystemWithProject(?x, true)</code>
4	The components with installation requirements should have at least one location point	<code>CbqComponent(?x)^(RelatingInstallRequirement(?x,true)^(RelatingLocationPoint≥ 1)(?x) → ComponentHasLocationPoint(?x, true)</code>
5	For model units of the same type, the relative position of the location point should be the same	<code>CbqModelUnit(?x)^(CbqModelUnit(?y)^(rdf:type(?x,z)^(RelatingLocationPoint(?x,m)^(RelatingLocationPoint(?y,n)^(rdf:type(?y,w)^(owl:equivalentClass(?z,w)^(owl:SameAs(?m,n) → HasConsistentLocation(?x,y)</code>
6	The components of the peripheral system should indicate the connection relationship	<code>CbqComponent(?x)^(Contains(?y,x)^(CbqPeripheralSystem(?y)^(ConnectWith≥ 1)(?x) → HasConnection(?x, true)</code>
7	The containment between parts and components should be indicated	<code>CbqPart(?x)^(Contains≥ 1)(?x,y) → HasAttachedComponent(?x, true)</code>
8	The model unit should have correct information detail	<code>CbqModelUnit(?x)^(RelatingInformationDepth(?x,RID)^(swrlb:greaterThan(?RID,N2) → HasCorrectInformationDepth(?x,y) [Taking information depth N2 as an example]</code>
9	Components should have the correct geometric representation accuracy	<code>CbqModelUnit(?x)^(RelatingGeometricAccuracy(?x,GA)^(swrlb:greaterThan(?GA,G2) → HasGeometricAccuracy(?x,y) [Taking information depth N2 as an example]</code>
10–14	Components of water supply and drainage system, HVAC system, electrical system, intelligent system and power system should have the correct connection relationship	<code>CbqComponent(?x)^(Contains(?y,x)^(CbqWaterSystem(?y)^(ConnectWith≥ 1)(?x) → HasConnection(?x, true)</code>
		<code>CbqComponent(?x)^(Contains(?y,x)^(CbqHvacSystem(?y)^(ConnectWith≥ 1)(?x) → HasConnection(?x, true)</code>
		<code>CbqComponent(?x)^(Contains(?y,x)^(CbqElectricSystem(?y)^(ConnectWith≥ 1)(?x) → HasConnection(?x, true)</code>
		<code>CbqComponent(?x)^(Contains(?y,x)^(CbqIntelligentSystem(?y)^(ConnectWith≥ 1)(?x) → HasConnection(?x, true)</code>
		<code>CbqComponent(?x)^(Contains(?y,x)^(CbqPowerSystem(?y)^(ConnectWith≥ 1)(?x) → HasConnection(?x, true)</code>
15	Components should not exceed their own space occupancy	<code>CbqComponent(?x)^(RelatingGeometricElement(?x,GE)^(RelatingSpaceOccupy(?x,SO)^(swrlb:greaterThan(?GE,SO) → HasConsistentBetweenGeomWithSpace(?x, true)</code>
16	Component with different materials should not overlap with each other	<code>CbqComponent(?x)^(CbqComponent(?y)^(RelatingMaterial(?x,mx)^(RelatingMaterial(?y,my)^(differentFrom(?mx,my)^(IntersectWith(?x,y) → HasInconsistentSpaceRelationship(?x, false)</code>
17	The property value of model units should be unique	<code>CbqModelUnit(?x)^(RelatingIdentifier = 1)(?x) → HasUniqueName(?x, true)</code>
18	Model units of the same type should have a consistent name	<code>CbqModelUnit(?x)^(CbqModelUnit(?y)^(rdf:type(?x,z)^(rdf:type(?y,w)^(owl:equivalentClass(?z,w)^(RelatingNameAttribute(?x,m)^(RelatingNameAttribute(?y,n)^(owl:SameAs(?m,n) → HasConsistentName(?x,y)</code>
19	Concrete components of different strengths should not overlap with each other	<code>CbqComponent(?x)^(CbqComponent(?y)^(RelatingMaterial(?x, concrete)^(RelatingMaterial(?y, concrete)^(RelatingStrength(?x,sx)^(RelatingStrength(?y,sy)^(differentFrom(?sx,sy)^(IntersectWith(?x,y) → HasInconsistentSpaceRelationship(?x,y)</code>
20	The secondary system should have the consistent color with its upper-level system	<code>CbqSystem(?x)^(CbqSystem(?y)^(Contains(?x,y)^(RelatingColor(?x,cx)^(RelatingColor(?y,cy)^(swrlb:equalTo(?cx,cy) → SystemHasConsistentColour(?y, true)</code>
21	The color of the components should be consistent with that of the system to which it belongs	<code>CbqComponent(?x)^(CbqSystem(?y)^(Contains(?y,x)^(RelatingColor(?x,cx)^(RelatingColor(?y,cy)^(swrlb:equalTo(?cx,cy) → ComponentHasConsistentColour(?x, true)</code>
22	The components of fire protection system should be red	<code>CbqComponent(?x)^(CbqFireProtectionSystem(?y)^(RelatingColorFeature(?x,z)^(swrlb:equal(?z, red)^(Contains(?y,x) → ComponentHasCorrectColor(?x, true)</code>
23	The project to which the component belongs to should be consistent with the project to which the system containing it belongs to	<code>CbqComponent(?x)^(CbqSystem(?y)^(CbqProject(?z)^(Contains(?z,y)^(Contains(?z,x) → HasConsistentProject(?x, true)</code>

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